

AmadorUAVs

Boreas



Amador Valley HS Unmanned Aerial Vehicles Team

2021-2022 AUVSI SUAS Technical Design Paper

Abstract:

This paper describes the technical approach that Amador Valley High School's Unmanned Aerial Vehicles Team (AmadorUAVs) took while designing their UAS, Boreas. The team has designed and fabricated a multi-rotor system from scratch that meets flight performance requirements. Additionally, the team has developed and integrated a software stack that allows Boreas to fly autonomously and complete mission tasks. As a first-time competitor with a tight budget, the team has carefully balanced design effectiveness and quality while maintaining a strict manufacturing budget. This paper overviews the design, manufacturing, and evaluation process for AmadorUAVs' 2022 flight system.



Created by the AmadorUAVs 2021-2022 Team:

Kalyan Sriram, Kai Gottschalk, Vincent Wang, Justin Park, Ishan Duriseti, Derick Mathews, Raghav Misra, Ethan Kuo, Kush Nayak, Angad Bhargav, Nikhil Sunkad, Karan Gupta, Lucas Zhong, Sahil Mehta, Jishnu Deep, Kathryn Xiong, Siddharth Bhargav, Aayush Gupta, Frank Yu, Akshay Javvaji, Tony Li, Dylan Kwong, Justin Zhao, Shantelle Tupaz, Adam Tout, Darsh Shah, Olivia Cheng.

1 Requirements & Acceptance Criteria

1.1 Introduction

SUAS requires autonomous flight, obstacle avoidance, telemetry streaming, object detection, mapping, and payload drop. The following analysis evaluates Boreas's ability to complete competition tasks and outlines team priorities given limited budget and time.

Requirement		Solution		
Criteria Section	Description	Design	Compliance	Confidence
1.2.1	Flight Performance	2.5.1	Fully Compliant	Confident
1.2.2	Radio Links	2.8	Fully Compliant	Very Confident
1.3	Path Planning	2.6.4	Fully Compliant	Very Confident
1.3	Obstacle Avoidance	2.7	Compliant	Very Confident
1.4	Collision Prevention	N/A	Not Implemented	N/A
1.5	Imaging & Object Detection	2.1, 2.2	Compliant	Confident
1.6	Air Drop	2.4	Compliant	Somewhat Confident

Requirements Compliance Matrix

1.2 Hardware Performance Requirements

1.2.1 Flight Requirements

The following flight performance requirements (Table 1) were estimated from mission rules (10 km waypoints + other tasks in < 30 minutes) and verified with simulation (see 4.3.1).

Specification	Required	Actual
Flight Distance	20 km [12 mi]	30 km [19 mi]
Flight Time	30 min	50 min
Average Velocity	15 m/s [34 mph]	15 m/s

Table 1: Flight performance requirements

1.2.2 Radio Link Requirements

The following radio links (Table 2) were identified to cover all mission requirements. Airfield size and estimated max distance from UAS to the GCS were measured to determine range requirements.

Usage	BW	Latency	Reliability
Telemetry	250kb/s	low	very high
Manual Control	100kb/s	very low	medium
Image Transport	50Mb/s	medium	high

Table 2: Radio link requirements

1.3 Path Planning

Accurate completion of flight tasks required plotting accurate and efficient waypoint plans for area survey while avoiding obstacles during both waypoint traversal and ODLC/mapping. The team determined a pre-generated path would be more reliable than real-time generation.

1.4 Collision Prevention

Competition rules allow multiple UAS flying simultaneously, risking collision. However, due to limited team resources and time, combined with the statistically low probability of collision, the team did not prioritize developing a collision avoidance system. A simpler precautionary avoidance system using Interop telemetry is being evaluated.

1.5 Image Capture and Object Detection

To complete ODLC and mapping, the drone must be able to take aerial images at greater than 100ft AGL (Above Ground Level). Flying at higher altitudes trades image resolution for mission time; simulated runs revealed a minimum survey altitude of 200ft was required to complete the mission within timeline. The minimum ODLC pixel width found to be feasible for both automatic labeling and manual classifications (through model accuracy testing) was about 45 pixels; 300ft was chosen as the standard survey altitude to provide the necessary resolution. High accuracy and low operator workload were required for autonomous and manual imaging systems.

1.6 Air Drop

The team set payload constraints while designing the Unmanned Ground Vehicle (UGV) and choosing Boreas's propulsion requirements. The UGV was kept to a maximum weight of 2kg (including water bottle) and the drone was required to be able to carry 3kg of payload.

2 System Design

2.1 Imaging System

2.1.1 Camera

An Imaging Source DFK-33UX183 industrial camera captures 20MP images (5472×3648 px) at 18FPS. Attached is a Computar V3522-MPZ 35mm fixed zoom machine vision lens (20.8° HFOV, 15.7° VFOV), allowing 160px of ODL horizontal resolution at 300ft. Preliminary testing was conducted to verify that the imaging system could resolve ODLs at this altitude (see **Figure 1**). A Python script on the onboard Jetson Nano (see **2.6.1**) captures images over USB. See also **2.2** and **2.8.3**.



Figure 1:
Sample ODL

A custom 3-axis brushless gimbal stabilizes and points the camera at regions of interest, irrespective of drone attitude, velocity, or vibration.

2.1.2 Survey Strategy

A "tiling" approach was implemented, in which the map was split into adjacent overlapping rectangles based on the configured altitude for the mapping task and camera specifications. A zigzag pattern is then applied to visit the center of each one of these rectangular tiles, where Boreas captures a photo of that individual tile. If tile centers are inside an obstacle or outside the geofence, a gimbal tilt at the closest point to that unreachable tile is performed to capture the image. See also **2.7**. Captured images are sent to ground via Wi-Fi link.

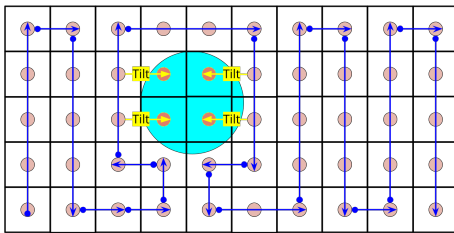


Figure 2: Survey Strategy

2.2 Object Detection

A YOLOv5 [6] object detection localization model, running on the drone's onboard Jetson Nano, identifies ODLs and crops out regions of interest, decreasing bandwidth and autonomous classification computing cost. See **3.1** for comparison to other models.



Localization

On the ground, the cropped image is passed through a Canny edge detector to eliminate colors and emphasize

edges. This is then run through further YOLOv5 models to classify orientation, shape, and character.



Figure 3: Further Classification

The original cropped and resized image is simultaneously run through two background-removal Convolutional Neural Networks (CNNs) that isolate the shape and character. A flood fill removes the remaining background pixels. The isolated shape and character are then color classified using an OpenCV most-prominent-color algorithm.



Shape and Letter BG removal; left to right: original, BG removal, floodfill

The final ODL is manually reviewed. Combining autonomous labels with manual review reduces bandwidth usage and human strain, while retaining the accuracy of manual classification.

2.3 Mapping

The same survey strategy as in **2.1.2** is employed. Captured images are geotagged, sent to a mapping server and accordingly rotated and projected on their respective positions on a full map. With high GPS accuracy and rectangular surveys, a high quality image can be created without complex stitching algorithms.

2.4 Air Drop

2.4.1 Unmanned Ground Vehicle Design

The UGV uses a treaded drive (4), with motors located on opposite corners for more compact packaging. An auxiliary plate provides vertical clearance for onboard drive electronics and a motorized winch. A winch cable runs through the center of the UGV and attaches to Boreas's servo drop system.

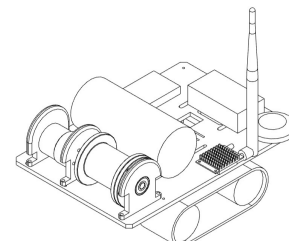


Figure 4: UGV CAD Drawing

2.4.2 UGV Control

A Pixhawk 4 flight controller running PX4 autopilot (in rover mode) provides UGV control in a small form factor. A dual H-bridge motor driver, along with an STM32F4 microcontroller, interfaces with the Pixhawk and enables direction and speed control of the 2 brushed DC motors. The team customized and tuned a mixer profile to integrate PX4's control stack with the UGV's differential drive. An RFD 915 MHz telemetry radio enables mission initiation, telemetry monitoring, and termination in event of system failure. *Pilot* (see 2.6.4) generates and uploads a UGV mission using Interop data, which is automatically executed after UGV deployment.

2.4.3 Drop System

Boreas descends to and hovers at 35 m for airdrop. A 3D printed winch and release mechanism lowers the vehicle from the UAS using a brushed DC motor, braided fishing line, and altitude estimates from the Pixhawk 4. Vertical velocity allows the STM32F4 to regulate winch RPM in closed loop. Gradually decreasing descent velocity optimizes drop time while preventing hard landing. When landing is detected ($v_{descent} \approx 0$, $\Delta h \approx 35$ m), a servo-actuated latch releases the line, allowing the UGV to move freely.

2.5 Aircraft

2.5.1 Airframe

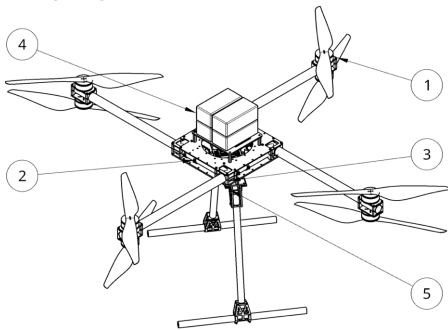


Figure 5: Boreas CAD Drawing

Multirotor airframes provide greater maneuverability and stability compared to other UAS. Boreas's coaxial octocopter configuration (1) allows for large propellers while minimizing airframe footprint and arm length. Carbon fiber tube arms are mounted along the edges of the milled carbon fiber plate (2), greatly increasing stiffness and creating space for electronics. Corner clamps (3), 3D printed from NylonX filament (see 3.4.3), sit between the carbon fiber body plates, and brace the four arms together, further increasing stiffness. Batteries are strapped to a top plate for easy access (4). The legs and payload are mounted to a laser-cut plywood modular

auxiliary plate (5) below the aircraft.

Empty Weight	3.5 kg [7.5 lb]
All Up Weight	11 kg [25 lb]
Flight Time	35 min
Hover Time	50 min
Airspeed	15 m/s [34 mph]
Airframe Diagonal (no props)	1391 mm [54.8 in]
Airframe Height	797 mm [31.4 in]
Propellers	24 in $\times$ 9 in

Table 3: Boreas Specifications

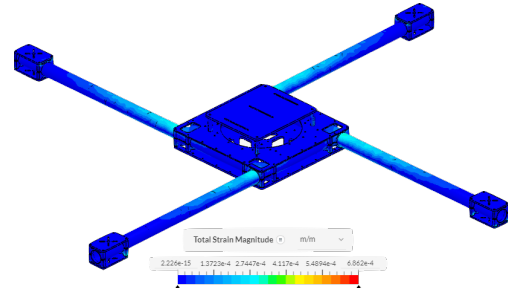


Figure 6: Boreas Total Strain Simulation

2.5.2 Propulsion & Power

Boreas uses eight XOAR Titan Air TA6012 130KV brushless motors, driven by Zubax Myxa field-oriented control (FOC) ESCs. Four 6S (22.2V) 10Ah lithium polymer batteries are wired in 2S2P, providing 20Ah @ 12S (44.4V). Power distribution blocks and switching regulators are mounted on a DIN rail on the auxiliary plate, providing power to ESCs and onboard electronics. The Myxa ESCs use commanded RPM setpoints (regulated by a PID loop), rather than conventional open loop ratiometric voltage setpoints. This maintains consistent thrust as battery voltage drops during flight, and provides a more predictable thrust curve ($F_{thrust} \propto RPM^2$), improving control performance (especially at high throttle).

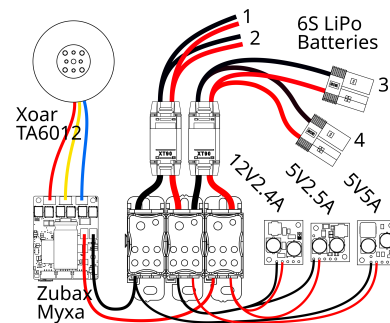


Figure 7: Boreas Propulsion & Power Diagram

2.6 Autopilot

2.6.1 Flight Controller

A Pixhawk 5X flight controller was chosen due to its performant processor (STM32F7), triple sensor and

power redundancy, vibration dampening, and sensor temperature stabilization, while being compliant with Pixhawk hardware standards. Latest generation sensors, expanded I/O, and a 100-Base-T Ethernet controller ensures the FC can meet any future tasks or UAS upgrades. A Jetson Nano SBC, chosen for low power draw, small footprint, and integrated GPU compute, enables the team to run ODLC region-of-interest identification models on the drone, significantly decreasing bandwidth use and offloading work from the GCS (2.2). The Jetson interfaces with the Pixhawk 5X over Ethernet.

QGroundControl (Figure 8) provides mission upload and monitoring.



Figure 8: QGroundControl

2.6.2 Peripherals

The Pixhawk 5X interfaces with the 8 Myxa ESCs using the UAVCANv0 protocol over the differential CAN transport. Differential signaling provides greater common-mode noise immunity than PWM or I²C and allows real-time RPM, power, and temperature feedback. Two CUAV C-RTK 9P (U-Blox ZED-F9P) GNSS modules provide centimeter-level localization with simultaneous connections to multiple GNSS constellations.

2.6.3 Flight Stack

Boreas is piloted by the PX4 open source autopilot and other parts of the Dronecode flight stack. See also 3.5. PX4 implements many autonomous control features including waypoint navigation, takeoff and landing, telemetry streaming, and geofencing, satisfying mission criteria. An Extended Kalman Filter (EKF) fuses data from IMUs, pressure sensors, and GNSS modules to generate an accurate attitude and heading estimate for Boreas. The MAVLink protocol facilitates communication between the autopilot and GCS. The Dronecode ecosystem also provides the open-source QGroundControl GCS and MAVSDK libraries for commanding PX4.

2.6.4 Pilot

The team's path planner, *Pilot*, fetches missions from Interop and constructs an effective flight plan, which is opened in QGroundControl, verified, uploaded to Boreas, and monitored in real-time. This approach was chosen over previous MAVSDK programmatic control for system simplicity and improved ability to catch errors and modify missions. Pilot's Python daemon streams telemetry to Interop at 1 Hz.

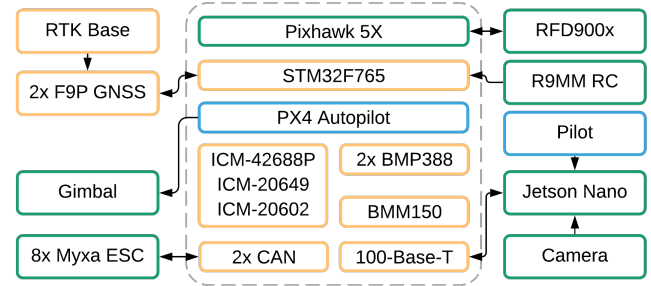


Figure 9: Toplevel Autopilot Component Diagram

2.7 Obstacle Avoidance

Pilot projects the flight area and obstacles onto a 1-meter 2D grid. The A* shortest-path algorithm[1] is applied, generating a path that hits all waypoints while routing around obstacles. Adjacent points are checked and deleted if redundant.

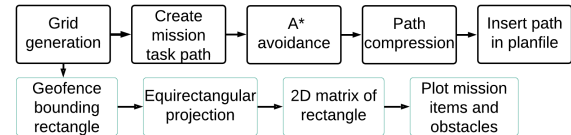


Figure 10: Obstacle Avoidance Flow

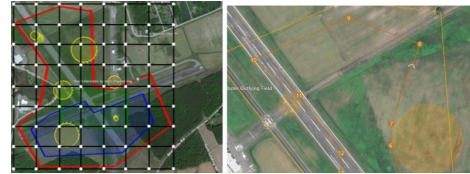


Figure 11: Grid approach, Avoided Obstacles

2.8 Communications

Multiple radio links are utilized (as mentioned in 1.2.2) to complete various mission objectives.

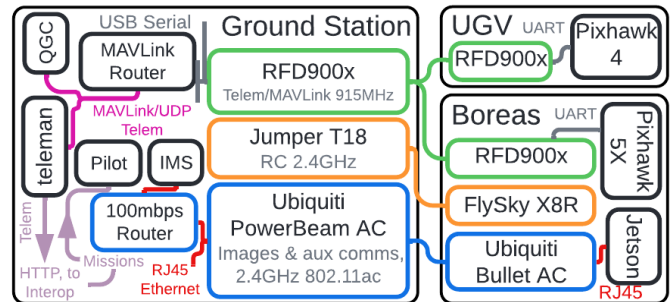


Figure 12: Radio Block Diagram

2.8.1 Manual (RC) Control

A Jumper T18 transmitter, paired with a FrSky X8R receiver, is used for manual override in the event of systems failure. The receiver utilizes frequency hopping 2.4GHz ACCST technology, and is wired to the Pixhawk 5X using S.BUS, providing reliable RC control.

2.8.2 Telemetry & Autonomous Control

Three 915 MHz RFDesign RFD900x long-range telemetry radios facilitate P2MP (Point to Multipoint)

MAVLink communication between GCS, rover, and drone. Max range of 10km with hardware-accelerated AES encryption, ensuring connection security. Used for autonomous control, mission upload, and telemetry streaming.

2.8.3 Image/Video Transfer

The 915MHz telemetry radio proved inadequate for imaging tasks due to limited bandwidth ($\approx 500\text{kbps}$). Hence, image/video transfer utilizes a 2.4 GHz WiFi system with a Ubiquiti PowerBeam directional antenna on the ground, connected to a Ubiquiti Bullet (with 12 dB_i antenna), providing a $\approx 100\text{mbps}$ link.

Images are transported over the WiFi link using the HTTP3/QUIC transport layer[5]. This transport uses UDP, eliminating TCP ack/retransmit overhead, perfect for an "unreliable" radio link. Additionally, QUIC supports a single roundtrip TLS handshake, compared to TCP's four roundtrip handshake.

2.8.4 Antenna Tracker

A two-axis antenna tracker (**Figure 14**) autonomously points the directional PowerBeam antenna (1) at Boreas. The design employs two NEMA 23 stepper motors (2), a 3D printed cycloidal drive (3), 4" turntable (4), and spur gears (5). The frame also features rigid carbon fiber counterweight tubes (6) and is mounted (7) on a speaker stand. See also 3.7.1.

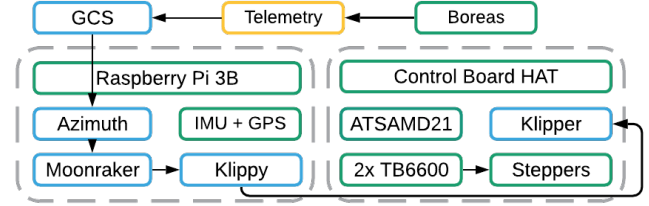


Figure 13: Antenna Tracker Control

A trajectory planner, running on a Raspberry Pi 3, receives MAVLink GNSS telemetry messages and emits GCode motion commands over CDC-ACM/USB. This is consumed by an ATSAMD21 microcontroller running the Klipper 3D printer firmware, which is assembled on a custom Pi HAT PCB. Klipper provides smooth acceleration control, an API (Moonraker), and a control UI (Fluidd). An IMU provides an absolute heading reference and compensation for unlevel ground. The resulting tracker can maintain a high-quality link to the drone within 2 degrees of accuracy.

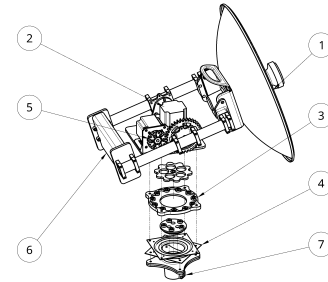


Figure 14: Antenna Tracker CAD Model

3 Alternatives Considered

3.1 Object Detection

During testing, the chosen YOLOv5 ODLC pinpointing model outperformed *EfficientDet-D0*[10] and conventional *OpenCV Simple Blob Detection* significantly in accuracy.

Method	mAP@0.5
YOLOv5	0.99
EfficientDet-D0	0.88
Blob Detection	0.95

Table 4: Relative Pinpointing Model Accuracy
The team is currently evaluating *Slicing Aided Hyper Inference* [7] to slice images into smaller chunks. This may improve accuracy on real-world images.

3.2 Mapping

The industry standard OpenDroneMap framework was considered for mapping. ODM supports generating geo-referenced, orthorectified images and is easy to integrate. However, benchmarks ($\approx 10\text{min}$ for 77 images) proved ODM to be too slow to generate a result within the mission demonstration timeline; a custom solution was implemented instead (2.3).

3.3 Air Drop

Multiple drop mechanism alternatives were considered:

- **Parachute:** low precision, unreliable
- **Mechanical brake:** complex, high risk of failure
- **Rheostatic brake:** insufficient brake power
- **Fishing reel:** unreliable, sadge

A powered descent was chosen because it avoided these issues. The winch mechanism was moved from the UAS to UGV, reducing risk of propeller interference and decreasing payload weight after release.

3.4 Aircraft

3.4.1 Airframe Layout

The conventional octocopter layout used previously lacked rigidity and space for electronics, but scaling up the existing design was unfeasible due to material cost and transportation limitations. Edge mounted coaxial arms (further braced by printed T clamps) significantly increased center space and frame rigidity. Minor efficiency loss of the coaxial layout [4] was outweighed by structural benefits.

3.4.2 Propulsion

Motor efficiency, thrust, and cost were considered when designing Boreas's new propulsion system. The XOAR TA6012 provided the highest efficiency within budget.

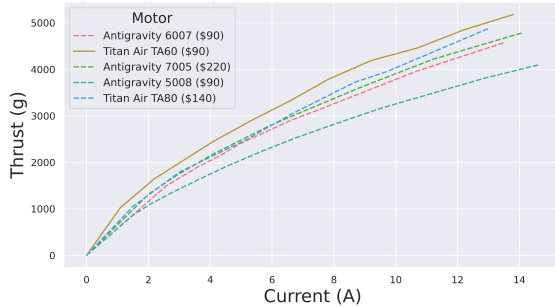


Figure 15: Comparison of Motor Efficiency

3.4.3 Materials

The frame clamps were originally designed to be aluminum, but CNC machining proved difficult, and simplifying would incur a large weight penalty. Another option considered was compression-molded carbon, which was inexpensive and very strong, but difficult to manufacture. 3D printed NylonX was chosen for ease of manufacturing while retaining high stiffness.

3.4.4 ESC Mounting

ESCs were previously mounted on the drone arms due to space constraints, which facilitated easy maintenance and shortened wiring, but was more fragile. The current design moves the ESCs to the center frame, providing multiple benefits:

- lighter & aerodynamic (durable plastic cases removed)
- significantly reduced dust/debris ingress
- increased protection from projectiles & crashes
- CAN cables protected from EMI or physical damage
- more aerodynamic airframe

Lower gauge, shielded cables were employed to prevent EMI issues arising from longer motor phase cables.

3.4.5 Power Distribution

A Matek X-Class PDB was used previously due to market availability and ease of use. However, with increased frame size and power requirements, it became impractical to add devices to the PDB, as it lacks a secondary battery connection. Instead, the team used a DIN rail based power distribution block system onto the auxiliary plate, which splits both 6S and 12S voltages to electronics on the drone. The distribution blocks are more modular, as they do not require solder, improving expandability and manufacturability. Despite the minor weight penalty, crimped connections prove highly reliable in harsh environments, and are immune to uneven heating and cold joints, unlike soldered connections.

3.5 Autopilot

3.5.1 Flight Stack

Table 5 lists the criteria the team used to choose a suitable autopilot. PX4 Autopilot was selected for Boreas.

Criteria	Ardupilot	PX4
Autonomous Flight	Good	Good
SITL simulation	Ok	Good
Community Support	Ok	Good
FOSS Licensing	GPLv3	BSD
Primary GCS	MP	QGC
Standards Compliance	Ok	Good
Aux. actuator control	Good	Ok
Flexibility	Ok	Good

Table 5: Autopilot Criteria

3.5.2 Pilot

B-spline curves[9] were evaluated to smooth flight trajectory and reduce flight time. However, simulation revealed no significant benefits from this approach.

3.6 Obstacle Avoidance

The A* pathfinding algorithm was compared to *artificial potential fields (APFs)*[8] and *evolutionary path planning models*[2]. However, genetic algorithms were slow and vulnerable to converging on locally optimal (but globally suboptimal) solutions. APFs increased complexity and suffered from local minima traps without yielding significant performance improvements over A*.

3.7 Communications

3.7.1 Antenna Tracker Drive

Multiple motor types were considered. Stepper motors met requirements best:

Drive Type	Precision	Continuous Torque	Simple	Low Cost	3D Printed
Standard Servo	●	○	●	●	●
Continuous Servo	●	●	●	●	●
Brushed DC	○	●	●	●	●
Brushless DC	●	●	●	○	○
Stepper	●	●	●	○	●
Spur Gear	●	●	●	●	●
Planetary Gear	●	●	●	○	○
Cycloidal Drive	●	●	●	●	●

● = met ○ = partially met ○ = not met

3.7.2 Antenna Tracker Control Firmware

Last year's custom tracker control firmware was replaced with Klipper due to the high engineering time required to implement more precise stepper timing and acceleration control.

3.8 Other

3.8.1 Filament

The team relies heavily on 3D printing to prototype and manufacture components. PLA was originally used for low cost and ease of printing, but was prone to

heat warping. For production, the team transitioned to stronger engineering-grade polymers. PETG (in-expensive, temp. resistant, non-ductile) was used in

non-critical components. Nylon composites were used on mission-critical parts for high ductility/shatter resistance and very high temperature resistance.

4 Testing & Evaluation Plan

4.1 Motivation

Rigorous procedures are followed to reduce the risk of costly failures and improve safety. The team analyzed past failure modes to prevent future issues.

4.2 Developmental Testing

4.2.1 Flight Time

Extensive mission tests were simulated to verify that Boreas can complete a 30-minute mission at its estimated cruise speed and chosen flight parameters. Typical performance timestamps are listed below.

Task	Time	Transit Time
Waypoints	00:05:14	00:05:35
Mapping	00:10:53	00:11:23
Area Survey	00:26:28	00:26:50
Air Drop	00:28:55	00:29:30 (RTH)

Table 6: Flight Time Timestamps

4.2.2 Environment/Weather Conditions

The team will perform full mission testing in a variety of weather conditions including heavy sun, wind, and light rain to gain insight into risks due to environmental factors. Risks include electronics overheating, part deformation, and decreased flight/control stability.

4.2.3 Air Drop

Air drop presents an especially high number of failure modes. The system will be tested in isolation (from a building or drone in hover) at least 20 times in addition to multiple full mission tests to minimize risk of failure. The mechanism is inspected after each drop for excessive wear, overheating or drive failure. Descent speed and landing accuracy are also measured and optimized.

4.2.4 WiFi Antenna & Tracker

The team plans to test the antenna tracker for at least 15 hours on dummy and real vehicle targets to ensure mechanical and control stability. During testing, the antenna tracker is inspected for issues such as mechanical play, vibrations, and interference that may disrupt the function of the antenna tracker. The Wi-Fi link will also be tested for range and reliability; speeds will be measured at a variety of distances and conditions.

4.2.5 Object Detection

Vision models were trained on thousands of auto-generated images and evaluated on similar test images, comparing multiple metrics including mAP (mean

average precision), recall, and precision. Models were then verified on the field using cardboard ODLC props to ensure satisfactory performance on real images.

Model	mAP@0.5	Recall	Precision	Time (s)
Pinpointing	0.99	0.983	0.976	0.6695
Character	0.994	0.983	0.976	0.4443
Shape	0.987	0.964	0.966	0.4462
Orientation	0.972	0.96	0.963	0.1750

Table 7: Comparison of Model Performance

4.2.6 Propulsion System Validation

Thrust output & current consumption (measured with thrust stand) were compared against manufacturer specs to ensure motor quality. Measured efficiency was lower than spec but within an acceptable error margin.

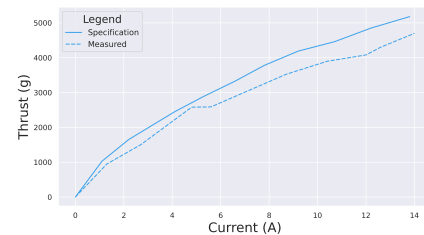


Figure 16: Motor Performance Validation

The team performed 3 hours of endurance testing to ensure each motor/propeller could tolerate high thrust for an extended time without mechanical failure.

4.2.7 Drive Tuning

FOC ESCs feature higher efficiency at the cost of increased complexity. Badly tuned parameters like power ramp limits can induce catastrophic drive malfunction. For example, the RPM PID controller was tuned and tested over a wide range of RPM setpoints to prevent runaway and ensure optimal reference tracking (Figure 17).

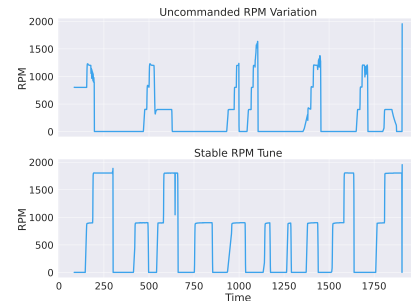


Figure 17: RPM Tuning

4.2.8 Vibration Profiling

Early design iterations had very high vibration, decreasing controller performance and introducing safety risks. Vibration logs were collected and analyzed in the

frequency domain (using FFTs) to diagnose the issue and improve airframe design. Sensor low pass and notch filters were configured to attenuate $60 \text{ Hz} \pm 15 \text{ Hz}$ motor vibration, significantly improving vibration characteristics. PX4's (beta) adaptive notch filter, which uses a real-time FFT to update notch frequency and bandwidth, was also evaluated, but did not perform consistently.



Figure 18: FFT, Default v. Tuned Filters

4.2.9 Controller Tuning

Default gains for PX4's PID controllers were inadequate, leading to overshoot and poor reference tracking. The team first employed the built-in auto-tune function, and then manually adjusted gains to further improve performance. The attitude, velocity, and position loops of PX4's cascading controller were tuned manually to improve position hold performance.

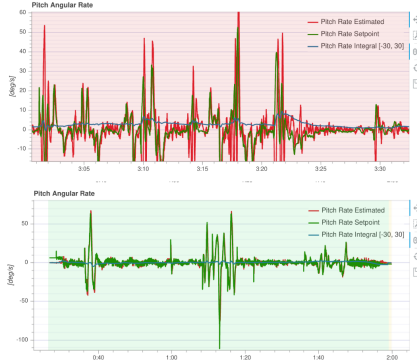


Figure 19: Untuned v. Tuned PID

4.3 Mission Testing

4.3.1 Software in the Loop (SITL) Simulation

The team created accurate mock mission environments in the *Gazebo* simulation framework to test *Pilot*, mapping algorithms, and ODLIC models. Simulated drone

telemetry was logged and analyzed:

Waypoints Hit	465/465
Mean Waypoint Error	0.4 m [1.31 ft]
Obstacles Avoided	30/30
Minimum Obstacle Clearance	5 m [16.4 ft]
Total Distance Traveled	50.5 km [31.3 mi]

Table 8: Pilot SITL Testing Results

The team plans to execute further simulated tests of the software stack, utilizing the Interop server's grading to accurately estimate total competition score.

4.3.2 Flight Testing Procedures

The flight stack was tested on mock missions created in QGC and inserted into Interop. *Pilot*-generated flight plans performed consistently on real mission tests. Flight logs were analyzed to identify under performing systems (i.e. power usage or controller tracking), airframe damage (i.e. increased vibration), and grade overall performance. Real-world flight characteristics and performance are compared to estimated characteristics (Table 3).

Waypoints Hit	64/64
Mean Waypoint Error	0.7 m [2.3 ft]
Obstacles Avoided	6/6
Minimum Obstacle Clearance	3.4 m [12.1 ft]
Total Distance Traveled	952 m [3123 ft]

Table 9: Pilot Real-World Results

The team plans to further run full-length missions at the local flight ground. All software and hardware components will undergo weekly mission flights to prove readiness. Success criteria for full mission readiness include:

- **Reliability:** Boreas must fly multiple 50-minute missions with no critical failures and zero failed obstacles.
- **Precision:** Waypoint error must be minimized. The ODLIC system should catch a majority of ground objects.
- **Safety:** All failsafes (5.2.4) must be tested in multiple missions.
- **Timeline:** Boreas must complete full-length (6mi) missions within the 30-minute deadline. Setup and tear-down must be reduced to 10 minutes each.

5 Safety, Risks, & Mitigations

5.1 Developmental Risks & Mitigations

5.1.1 Safe Equipment Operation

UAS development involves the use of hazardous tools like soldering irons, drills, and saws. Team members exercise the SAFE method for tool operation: Supervision, Assessment of work area (and PPE), Fitness/skills (member training), and Equipment (Good condition,

safe use).

5.1.2 Hazardous Materials

A fire blanket and extinguisher are mandated at all meetings in case of lithium-polymer battery fire. Batteries are routinely checked for damage; damaged or puffed batteries are sealed and disposed of safely. PPE (N95 masks, goggles, hearing protection) is worn when han-

dling toxic materials (like carbon fiber dust) or operating tools.

5.1.3 Clearance & Flight Test Procedures

To minimize risk to bystanders, flight tests are conducted at a remote area. Arming is not permitted if people are within 10 feet of Boreas. Additionally, a geofence restricts the drone to within the flight zone.

5.1.4 Electrical Safety Procedures

All batteries and electronics must be disconnected before handling. All connectors are labeled and new connections confirmed by a trained member before plugging in. All tools are insulated. Electrostatic discharge mitigation protocols are practiced when handling delicate circuits.

5.2 Mission Risks & Mitigations

5.2.1 Human Control Operation Failure

To minimize risk of operator error, only experienced team members act as safety pilots. Autonomous maneuvers are cross-checked before upload. Flight termination requires two simultaneous switches to prevent accidental activation.

5.2.2 Airframe Damage

All airframe components, especially 3D printed parts, are checked for damage before flight. All fasteners are assembled with threadlocker and routinely inspected.

5.2.3 Autopilot/Control Failure

In case of autopilot failure, the safety pilot takes over control, overriding autonomous systems. The safety pilot can also terminate flight in case of emergency.

5.2.4 Autopilot Failsafes

All built-in PX4 failsafes were configured (Table 10) and tested, both in SITL and on the field, by inducing failures manually (i.e. unplugging radios) and automatically using MAVSDK's failure injection API.

Description	Configuration
Low Battery	RTH@25%, Land@20%
RC Loss	RTH@60 s
Telem Loss	RTH@30 s, Term@180 s
Geofence Violation	Hold
GNSS Loss	Land

Table 10: Failsafe Configurations

5.2.5 Redundancy

Hardware and software redundancy has been implemented on many aircraft systems to reduce severity in case of component failure:

- redundant IMUs on flight controller
- redundant power supplies for flight controller
- coaxial layout tolerates drive failure (up to 1 per arm)
- telemetry radio diversity (2 antennas)
- redundant RTK GNSS modules

References

- [1] Bertram Raphael Peter E. Hart Nils J. Nilsson. "A Formal Basis for the Heuristic Determination of Minimum Cost Paths". In: 1968.
- [2] Gihan Nagib and Wahied Gharieb. "Path planning for a mobile robot using genetic algorithms". In: Oct. 2004, pp. 185–189. ISBN: 0-7803-8575-6. DOI: 10.1109/ICEEC.2004.1374415.
- [3] Adam Bondyra et al. "Performance of Coaxial Propulsion in Design of Multi-rotor UAVs". In: *AUTOMATION*. 2016.
- [4] Bart Theys et al. "Influence of propeller configuration on propulsion system efficiency of multi-rotor Unmanned Aerial Vehicles". In: June 2016, pp. 195–201. DOI: 10.1109/ICUAS.2016.7502520.
- [5] Adam Langley et al. "The QUIC Transport Protocol: Design and Internet-Scale Deployment". In: 2017.
- [6] Jacob Solawetz Joseph Nelson. *YOLOv5 is Here: State-of-the-Art Object Detection at 140 FPS*. [Online; accessed 19-July-2020]. 2020. URL: <https://blog.roboflow.com/yolov5-is-here/>.
- [7] Fatih Cagatay Akyon et al. *SAHI: A lightweight vision library for performing large scale object detection and instance segmentation*. Nov. 2021. DOI: 10.5281/zenodo.5718950. URL: <https://doi.org/10.5281/zenodo.5718950>.
- [8] Iswanto1 et al. "Artificial Potential Field Algorithm Implementation for Quadrotor Path Planning". In.
- [9] Apoorva Joshi. *Constructing a cubic Bezier that passes through four points*. URL: <https://apoorvaj.io/cubic-bezier-through-four-points/>.
- [10] Quoc V. Le Mingxing Tan Ruoming Pang. *EfficientDet: Scalable and Efficient Object Detection*. URL: <https://arxiv.org/pdf/1911.09070.pdf>.